

What's New In WIRELESS CHARGING

Wireless charging technology will give engineers the freedom to make devices smaller and lighter by eliminating the energy storage devices, such as bulky batteries, in them.

The improvement in processing capabilities of our handheld devices makes wireless charging a convenient or rather an essential technology

SHARAD BHOWMICK

Wireless charging, a technology that was demonstrated over a century ago by Nikola Tesla but could not find any practical use for a long time is now becoming ubiquitous in a wide variety of devices, such as smartphones and smartwatches. A considerable amount of research and development is being done to improve the capabilities and even introduce wireless charging in highly power-intensive applications like electric vehicle (EV) charging. Wireless charging is of three types: resonant charging, inductive charging, and RF charging.

Wireless charging is a very hot topic at present; the wireless charging market is expected to experience a growth of 26%

CAGR from 2022 to 2027. The growth is reflected in the advancement in technology and the increase in the number of new launches. The latest trends in wireless charging modules are:

- Adoption of Qi 1.3
- Integration of antenna board
- More powerful processors
- Better protection features
- Higher power transfer rate
- Increased interfaces
- Smaller modules for smarter wearables

Adoption of Qi 1.3

Qi 1.3 is the latest update from WPC, rolled out in the middle of 2021. This is a major update since the launch of Qi 1.2 in



Fig. 1: Wireless charging of a mobile phone
(Source: Link)